

Chengbo Zhang

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Research interests: Urban Transportation Systems, Human Mobility, Spatial Networks, Agentic AI, LLM

Education

PhD in Transportation Engineering

McGill University

2026 – Present
Montreal, Canada

- Fully funded by Excellent Graduate Scholarship
- Supervisor: Prof. Jiangbo Yu

Master in Urban Planning

Harbin Institute of Technology (Shenzhen)

2023 – 2025
Shenzhen, China

- Awarded the First-class Academic Scholarship
- Dissertation: “Dynamic community structure and flow inference in on-demand delivery network”

B.E. in Urban Planning

Xiamen University

2018 – 2023
Xiamen, China

- Awarded the Merit Student honor for three consecutive years
- Awarded the National Encouragement Scholarships (equivalent to first-class scholarships)

Publications

Peer-reviewed journal articles

- **Zhang, C.**, Cui, Q., Ma, X., Li, Y., Xiao, Z., & Yu, J.* (in press). Introducing spatiotemporal order crowdedness into the on-demand delivery management data pipeline: A dynamic inverted two-step floating catchment area (Di2SFCA) method. *Journal of Transport Geography*.
- Ma, X., Zhao, Z., **Zhang, C.**, Liu, D., Cui, Q., Zhang, Y., Gong, Y.*, & Liu, Y. (2026). Understanding cyclist route choice: A hierarchical adversarial inverse reinforcement learning approach to behavioral profiling. *Journal of Transport Geography*, Article 104779. [DOI]
- **Zhang, C.*** (2026). Visualizing spatiotemporal flows of on-demand deliveries: A 3D interactive animation of path-reconstructed rider trajectories in Beijing. *Environment and Planning B: Urban Analytics and City Science*, 53(6), 1411–1414. [DOI]
- **Zhang, C.**, Xiao, Z.*, Li, Y., Wang, C., Zhang, W., & Hong, W. (2026). Uncovering the community structure and evolutionary dynamics of on-demand instant delivery networks. *npj Sustainable Mobility and Transport*, 3, Article 26. [DOI]
- **Zhang, C.**, Tang, X., Ma, X., Zhong, S., Zhao, B., Xiao, Z.*, & Gong, Y. (2026). Crafting a walkable and rideable urban green space system: Human-centered park network planning supported by active leisure travel trajectories. *Urban Forestry & Urban Greening*, 117, Article 129235. [DOI]
- Xiao, Z., Chen, X., **Zhang, C.***, & Gong, Y. (2025). Effects of road greenery and sky openness on non-commuting cycling route choice: Insights from online map navigation. *Transportation*. Advance online publication. [DOI] (corresponding author; supervisor as first author).
- **Zhang, C.**, Yang, X., Huang, J., & Xiao, Z.* (2025). Uncovering spatial patterns of environmental influence on the paces of active leisure travel. *Cities*, 162, Article 105971. [DOI]
- Xiao, Z., Li, Y.*, & **Zhang, C.** (2025). Impact of spatial function complementarity on outshopping flows: A spatial interaction model. *Travel Behaviour and Society*, 39, Article 100965. [DOI]
- Yang, X., **Zhang, C.***, & Yang, L. (2025). Revealing spatial patterns and environmental influences on jogging volume and speed: Insights from crowd-sourced GPS trajectory data and random forest. *ISPRS International Journal of Geo-Information*, 14(2), Article 80. [DOI]
- **Zhang, C.**, Shi, D., & Xiao, Z.* (2024). Integrating variable importance and spatial heterogeneity to reveal the environmental effects on outdoor jogging. *Computational Urban Science*, 4(1), Article 45. [DOI]

- Xiao, Z., **Zhang, C.***, Li, Y., & Chen, Y. (2024). Community park visits determined by the interactions between built environment attributes: An explainable machine learning method. *Applied Geography*, 172, Article 103423. [DOI] (corresponding author; supervisor as first author).

Peer-reviewed conference proceeding

- **Zhang, C.**, Zhang, L., Liang, W., & Yang, X. (2025). Applying Infomap-based hierarchical community detection for multi-level city on-demand delivery management. In *Proceedings of the 2025 28th International Conference on Computer Supported Cooperative Work in Design (CSCWD)* (pp. 2629–2634). IEEE. [DOI]

Preprints

- Yu, J., Eslami, A., Wang, J., Wang, Y., **Zhang, C.**, & Paquette-Greenbaum, S. (2026). Multimodal large language model agents across transportation systems lifecycle: Early-stage progress review and policy implications. *SSRN*. [Preprint]
- **Zhang, C.**, & Xiao, Z. (2026). Large language models as delivery rider: Generating instant food delivery riders' routing decision with LLM agent framework. *arXiv*. [Preprint] Extended journal version in preparation.
- **Zhang, C.**, & Xiao, Z. (2025). Deciphering delivery mobility: A city-scale, path-reconstructed trajectory dataset of instant delivery riders. *arXiv*. [Preprint]

* Corresponding author.

Conference Presentations

- Decoding Spatiotemporal Order Crowdedness in Online-to-Offline Delivery: A Dynamic Inverted Two-Step Floating Catchment Area (Di2SFCA) Method. *International Time Geography Conference*. Toronto, Canada, July 2026.
- Large Language Models as Delivery Rider: Generating Instant Food Delivery Riders' Routing Decision with LLM Agent Framework. *The 3rd International Conference on Urban Science and Intelligence*. Bali, Indonesia, July 2025.
- Dynamic network of instant delivery: Multi-layer detection, temporal evolution, and border thickness. *International Symposium on Spatiotemporal Data Science, Harvard Center for Geographic Analysis Conference*. Boston, America, May 2025.
- Uncover the Dynamic Community Structure and Evolutionary Dynamics of Instant Delivery Network. *Annual Meeting of the American Association of Geographers*. Detroit, America, March 2025.
- Dynamical Community Detection and Spatiotemporal Analysis based on Multilayer Instant Delivery Network. *European Transportation Conference*. Antwerp, Belgium, September 2024.
- Examine the nonlinear and interactive effects of built environment on community park visits. *National Forum on Computational Social Science*. CUHK Shenzhen, China, November 2023.

Academic Awards

- **Runner up Award** (Top 2 of 10), International Time Geography Conference Graduate Student Paper Competition, 2026.
- **Outstanding Poster Award** (Top 8 of 32), The 3rd International Conference on Urban Science and Intelligence, 2025 (evaluated by Prof. Marta Gonzalez, Prof. Carlo Ratti, Prof. Fengli Xu, and Prof. Yong Li).
- **Second Place Prize** (Top 2 of 8), AAG Annual Meeting John Odland SAM Student Paper Competition, 2025.
- **First Prize** (Top 3 of 100+ teams), Chengyuan Cup Planning Decision Support Model Design International Competition, 2024 (as lead contributor).

- **First Prize** (Top 1%), Geoscene GIS Development Competition, 2024 (as lead contributor).
- **First Prize** (Top 10%), National College Student Statistical Modeling Competition – Guangdong Division, 2024 (as lead contributor).
- **Third Prize** (Top 5%), National College Student Statistical Modeling Competition – National Competition, 2024 (as lead contributor).
- **Top 100 Best Prize**, City Design track at the Digital China Innovation Contest, 2022 (as lead contributor).
- **Nomination Prize** (Top 10%), WUPENiCity International Urban Design Competition, 2022 (as lead contributor).
- **Excellent Award**, Xiamen Big Data Security and Open Innovation Application Contest, 2021 (as lead contributor).

Work Experience

Research Assistant	May 2024 – Dec. 2024
<i>Peking University Shenzhen Graduate School</i>	Shenzhen, China
• Utilizing heterogeneous graph learning to predict human mobility flows between urban units.	
Data Analysis Intern	Dec. 2022 – Feb. 2023
<i>CitoryTech Company (citory.tech)</i>	Suzhou, China
• Analyzing Dazhong Dianping POI and street view image data to assess travel attractiveness of attractions.	
Intern Planner	Jul. 2022 – Nov. 2022
<i>Tsinghua Tongheng Urban Planning & Design Institute (THUPDI)</i>	Beijing, China
• Processing mobile signal data in the Smart City Center to depict and analyze the commuting structure in Beijing.	

Skills

- Agentic AI: OpenAI API, Claude, Claude Code CLI.
- Machine learning and deep learning with Python (NumPy, pandas, GeoPandas, matplotlib, scikit-learn, SHAP, NetworkX, PyTorch, DGL).
- Statistical modeling: R, Stata, SQL, SPSS.
- Spatial analysis and visualization: QGIS, ArcGIS, Adobe Creative Suite, Microsoft Office Suite.
- Language: Fluent in English (IELTS 7.0, Duolingo 130).